

5 Organic matter as the foundation of root zone function

Organic matter links physical soil condition water nutrient storage microbial activity and root health.

What organic matter can do

Pathway	Main benefit	Important limit
Physical	Supports aggregation pore space infiltration water storage and resistance to crusting.	It cannot by itself remove a hardpan or provide an outlet for perched water.
Chemical	Adds nutrient-holding sites buffers pH change and stores nutrients in organic forms.	Release is gradual and variable; nutrient content must be counted.
Biological	Provides food and habitat for decomposers beneficial organisms and roots.	Activity falls when soil is too dry saturated saline cold or otherwise hostile.
Water and salt	Surface mulch reduces evaporation and can improve infiltration and moisture stability.	Some composts and manures add EC sodium chloride or ammonium.
Root health	Coarse mulch can encourage feeder roots and support organisms antagonistic to avocado root rot.	Mulch must be kept away from trunks and cannot replace irrigation or disease management.

Organic materials are not interchangeable

Material	Best viewed as	Questions to ask
Stable soil organic matter	Long-lived soil carbon and nutrient-holding capacity	Is the trend stable increasing or declining and how was it sampled?
Coarse surface mulch	Cover moisture protection carbon input and root habitat	Is it clean coarse kept off trunks and replenished as it breaks down?
Finished compost	Conditioner plus a measurable nutrient and salt source	Maturity stability C to N ratio EC pH sodium chloride nutrient analysis
Manure or manure compost	Nutrient-rich amendment with possible salt and ammonium risk	Source treatment analysis application rate food-safety and runoff concerns
Fresh residues	Short-term microbial food that may release or tie up nitrogen	C to N ratio particle size placement moisture and decomposition rate
Concentrated carbon or microbial product	Formulation-specific carbon compounds or selected organisms	Active content or strain viability storage dose habitat evidence and cost

6 Organic matter field use and quality checks

Use organic inputs as measured tools, not as automatic remedies.

How organic matter connects to the requested soil factors

- **pH.** Stable organic matter buffers rapid change. Decomposition creates organic acids, but whole-soil pH change is often modest and depends on buffering by clay and carbonates.
- **Microbes and chelation.** Carbon feeds organisms that release enzymes, organic acids, and metal-binding compounds. The effect is strongest close to decomposing material and active roots.
- **PGPR and root growth.** Roots and carbon inputs create habitat, but a named inoculant still needs crop- and site-specific evidence.
- **Texture and nutrient retention.** Organic matter adds holding capacity in sandy soil and helps aggregation in finer soil. It does not change the mineral texture class.
- **Compaction and infiltration.** Roots, organisms, and stable aggregates can improve pores over time. Severe restrictive layers may require separate diagnosis and remediation.
- **Water availability and aeration.** Better structure can increase water storage while preserving air-filled pores. Fine material or over-irrigation can still keep soil too wet.
- **Leaching and salts.** Improved infiltration can support leaching, but salty amendments add to the load. Drainage must be adequate.

Before applying compost manure or another carbon product

Check	Why it matters	Record
Maturity and stability	Unfinished material can consume oxygen heat roots or temporarily tie up nitrogen.	Test method date smell temperature and supplier specification
C to N ratio	Helps predict rapid N release or short-term N tie-up.	Total carbon total nitrogen and ratio
EC and salt ions	Avocado roots are salt sensitive and inputs may add sodium or chloride.	EC sodium chloride and application rate
Nutrients and ammonium	The input is part of the nutrient budget and concentrated ammonium can injure roots.	Total and available N P K Ca Mg S and ammonium when relevant
Physical quality	Particle size influences surface cover decomposition and oxygen movement.	Particle size moisture contaminants and placement
Rate and area	Tons per acre can hide a much higher local dose beneath individual trees.	Dry weight treated area depth and distance from trunk

Management principle

Organic matter may improve nutrient supply, but it does not remove the need to measure water quality, irrigation uniformity, salinity, pH, drainage, and root condition.